

Validating the Corrected Phase 1 Energy at High N

The energy-discretization fix plus a moderate λ resolves the “runt” cell of report 01, so Phase 2 refinement *decreases* perimeter instead of crashing — with the corrected well, seeded initialization becomes mandatory

July 2026

Status — measured: the fix is validated end-to-end at $N=30$ and $N=100$ on the torus

This report **validates** the double-well discretization fix explained in `docs/reference/phase1_energy_discretization_bug.md` and derived in `docs/math/06-phase1-energy-discretization/`. It is the post-fix companion to report 01-winner-take-all-partition-gap/, which characterised the high- N “runt” failure *under the buggy energy*. The numbers come from five completed runs (three $N=100$, two $N=30$); the reconstruction is anchored on the production run’s worst-cell area deviation and final perimeter (below).

Provenance

Date	2026-07-09 (runs of 2026-06-29 and 2026-07-07 ... 07-09)
Surface	torus $T(R=1.0, r=0.6)$, seed 84172851 throughout
Fix commit	6ff71a0 (interface_vec = u*(1-u)), branch fix/phase1-energy-discretization
Source runs	$N=100$: buggy $\lambda=2.1$ run_20260629_141012 (report 01 anchor, 5 levels, 348×328); corrected $\lambda=2.1$ run_20260707_192154 (3 levels, 224×212); corrected $\lambda=5.1$ run_20260709_081548 (<i>production</i> , 5 levels, 348×328 , $V=114,144$). $N=30$ ($\lambda=2.1$): random init run_20260707_002828; seeded init run_20260707_080824.
Producing script	make_figures.py (beside this report; run from repo root)
Method inputs	final_solution/ density fields (worst-cell + distributions); the production Phase 2 campaign’s iteration*.h5 (final_perimeter, constraint_violations)
Versions	Python 3.12.13, numpy 2.4.4, scipy 1.17.1, matplotlib 3.10.8, h5py 3.16.0
Anchors	production worst-cell area deviation = 0.0078 (0.8%, matching metadata.yaml worst_rel.dev); production Phase 2 final perimeter = 185.2546 (matching iteration_020 final_perimeter); extracted Phase 1 perimeter = 214.3413

Abstract

Status: measured. Report 01 traced the high- N Phase 2 failure to a “runt” cell manufactured during Phase 1 relaxation, but its measurements were taken under an energy whose double well was later found to be *mis-discretized* (coded as $u^2(1-u)^2$ rather than $q = u(1-u)$, making the well $\sim 25\times$ too weak; `docs/reference/phase1_energy_discretization_bug.md`). This report re-runs the high- N case on the *corrected* energy. Three findings. **(1)** Correcting the well *alone* does not cure the runt: at the original $\lambda=2.1$ the $N=100$ worst-cell area error

is 21.8%, essentially unchanged from the buggy 22.5% — because the mis-scaled well had also made λ an inert lever. **(2)** The correction *restores* λ as an effective knob: a moderate $\lambda=5.1$ drives the worst cell to 0.8% (zero cells beyond the 5% gate), after which Phase 2 refinement *decreases* perimeter ($214.34 \rightarrow 185.25$, -13.6%) and stays feasible (equal-area violation $\leq 3 \times 10^{-6}$) — where the buggy energy made perimeter rise ($217 \rightarrow 229$) and crash on infeasibility. **(3)** The steeper corrected well introduces a new requirement: at $N=30$, random initialization now *traps* in the symmetric diffuse state (43% worst cell, 23 imbalanced cells) while seeded initialization stays clean (0.7%). The operating point is *corrected energy + seeded init + moderate λ* .

1 Question

Report 01 (01-winner-take-all-partition-gap/) showed that on the torus at $N=100$, Phase 1 satisfies the *continuous* equal-area constraint yet the *discrete* winner-take-all partition contains one grossly undersized “runt” cell, which makes Phase 2 perimeter refinement infeasible (perimeter rises, then “local infeasibility”). Every measurement there was taken with the Phase 1 energy’s double-well term coded as $\varepsilon^{-1}(u^2(1-u)^2)^\top M(u^2(1-u)^2)$. That term was subsequently found to be a mis-transcription of the paper: the intended discretization is $\varepsilon^{-1}q^\top Mq$ with $q = u(1-u)$, so the coded well was $\int u^4(1-u)^4$ — eighth-degree and $\sim 25\times$ too weak — with a gradient inconsistent with it (full audit: docs/reference/phase1-energy-discretization_bug.md; derivation of the corrected form: docs/math/06-phase1-energy-discretization/).

The question this report answers: *on the corrected energy, does the runt resolve, does Phase 2 then converge, and what re-tuning does the $\sim 25\times$ scale shift require?*

2 Method

All runs share the torus $T(1.0, 0.6)$, seed 84172851, and the corrected energy (except the report-01 buggy anchor, kept for contrast). Two families:

- **N=100 runt sweep.** Three runs: the buggy $\lambda=2.1$ anchor from report 01; the corrected energy at the *same* $\lambda=2.1$ (isolating the well fix); and the corrected energy at $\lambda=5.1$ (the production run). For each, the discrete territory of the final **solution/** is the winner-take-all lumped-mass area $a_k = \sum_{i: \arg \max_{k'} u_{k'}(x_i)=k} v_i$ (v the lumped P_1 mass); we report the worst-cell error $\max_k |a_k - \bar{A}|/\bar{A}$ and the full sorted per-cell distribution.
- **N=30 init head-to-head.** Two corrected-energy runs identical but for `init.method` (`random` vs `seeded`), $\lambda=2.1$, isolating the initialization as the only variable.

For the production run we additionally read the Phase 2 campaign’s `iteration_*.h5`: the `final_perimeter` attribute (the per-iteration perimeter) and the `constraint_violations` array in `optimization_info` (the per-iteration equal-area feasibility), prepending the extracted Phase 1 perimeter 214.3413 as iteration 0.

Anchors. The production worst-cell reconstruction gives 0.0078, matching the run’s `metadata.yaml` `area_imbalance.worst_rel_dev`; the Phase 2 final perimeter 185.2546 matches `iteration_020`. These confirm the mesh rebuild and reshape before any conclusion is drawn.

3 Results

3.1 The runt resolves — but only with the energy fix *and* a moderate λ

Table 1 and Fig. 1 give the central result. Correcting the well at the original $\lambda=2.1$ barely moves the runt ($22.5\% \rightarrow 21.8\%$): the fix alone is *not* sufficient. The reason is that the $\sim 25\times$

mis-scaling had also swamped the crispness penalty, leaving λ unable to act (report 01 swept $\lambda \in \{1 \dots 10\}$ with no effect). On the corrected well λ is restored as a lever, and a moderate $\lambda=5.1$ equalizes all 100 cells — worst cell 0.8%, zero cells past the 5% gate.

N=100, seed 84172851	buggy well, $\lambda=2.1$	corrected, $\lambda=2.1$	corrected, $\lambda=5.1$
finest mesh (V)	348×328 (114,144)	224×212 (47,488)	348×328 (114,144)
worst-cell area error	22.5%	21.8%	0.8%
cells beyond 5% gate	1	1	0
Phase 2 outcome	perimeter rises, crash	—	−13.6% , converges

Table 1: The runt is cured by the energy fix *plus* a moderate λ , not by either alone. Middle column: correcting the well at the old λ leaves the runt essentially unchanged; the cure needs λ raised into its now-effective range.

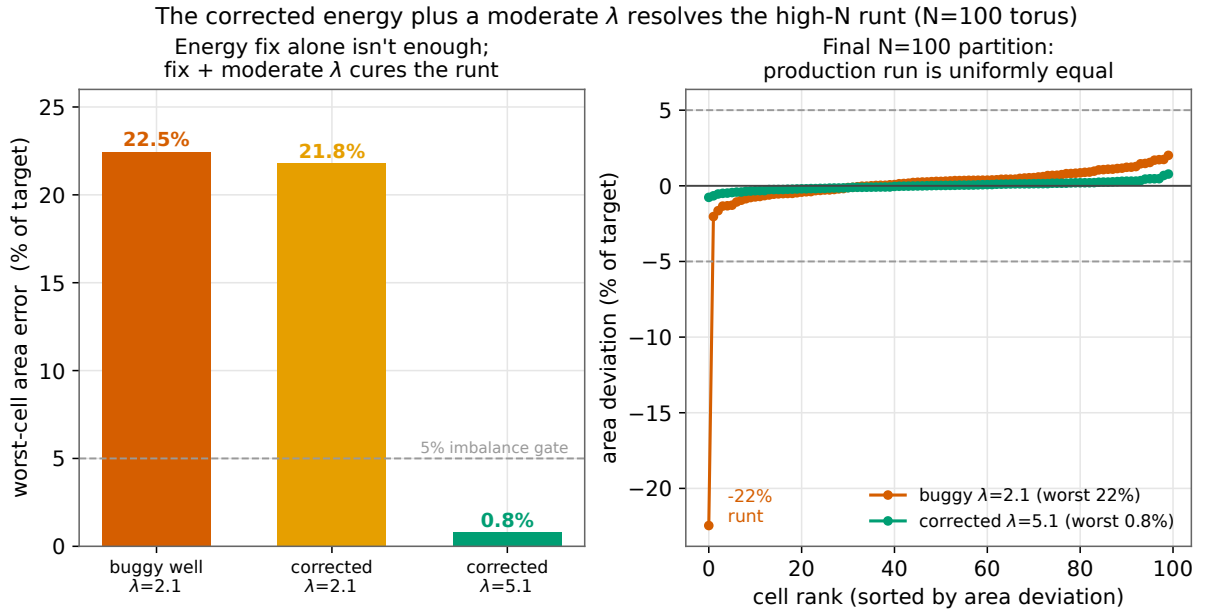


Figure 1: *Left*: worst-cell area error for the three $N=100$ settings — the energy fix alone (orange) is not enough, but fix + moderate λ (green) clears the 5% gate. *Right*: the final sorted per-cell area deviation — the production run (green) is uniformly equal, versus the single -22% runt of the buggy run (vermillion).

3.2 Phase 2 now decreases perimeter and stays feasible

On the production partition, Phase 2 behaves as intended (Fig. 2): the total perimeter falls from the extracted 214.34 to 185.25 (-13.6%) and the equal-area constraint holds throughout (max violation $\leq 3 \times 10^{-6}$; 4×10^{-10} at iteration 1). This is the exact opposite of the buggy runs, where the iteration-0 area violation was 0.05–0.16 and never resolved, so the feasibility-first solver lengthened perimeter ($217 \rightarrow 229$) chasing an unreachable feasible point. The per-iteration sawtooth in perimeter is the migration subsystem re-inserting variable points between solves; each solve drives the perimeter back down, and the best iterate (iteration 20) is the exported production partition.

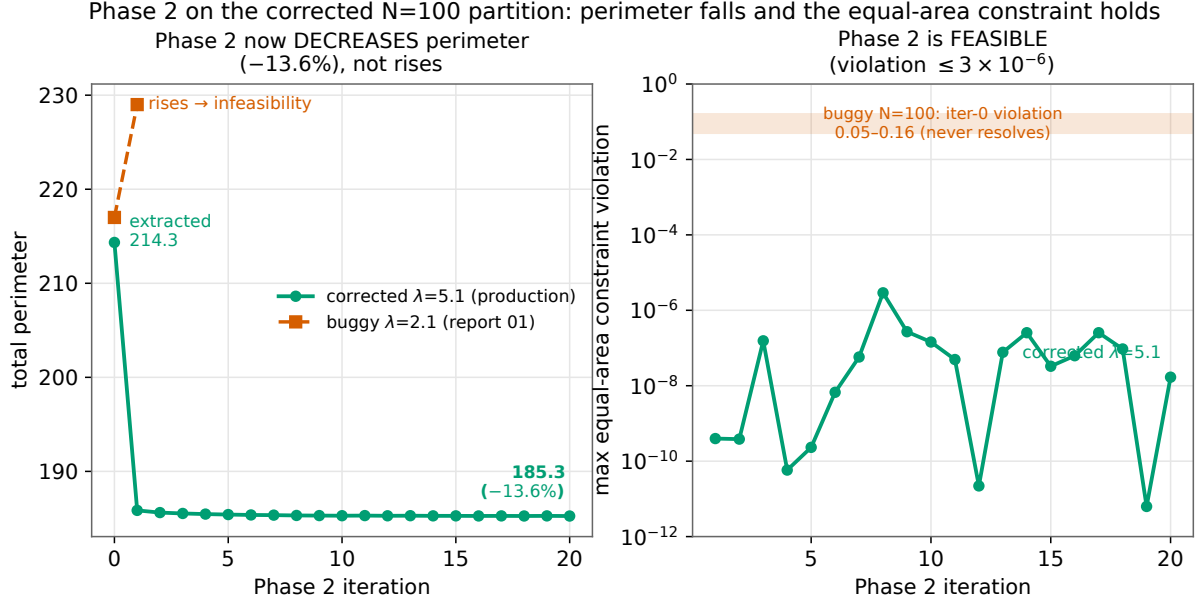


Figure 2: Phase 2 on the corrected $N=100$ partition. *Left*: perimeter *decreases* (-13.6% , green), versus the buggy run’s rise to infeasibility (vermilion, from report 01). *Right*: the equal-area constraint violation stays at $\sim 10^{-6}$ or below (green) — the buggy runs sat in the shaded 0.05–0.16 band and never left it.

3.3 The corrected well makes seeded initialization mandatory

The $\sim 25\times$ -steeper well has a side effect (Fig. 3): the *symmetric diffuse state* (every cell at density $1/N$ everywhere) becomes a genuine local minimum, where the projected gradient damps perturbations rather than amplifying them. **Random** initialization sits essentially at that state and freezes: at $N=30$, $\lambda=2.1$, it ends with 23 cells past the 5% gate and a 43% worst cell. **Seeded** (farthest-point Voronoi) init, which hands every cell a contiguous region from iteration 0, ends clean (0.7%, zero imbalanced) from the identical config. Under the old flatter well this trap was weak, so random “worked” at low N ; on the corrected energy it does not. Random init can still escape if λ is pushed hard (a level-0 sweep found escape at $\lambda \gtrsim 10$), but large λ over-crisps and costs perimeter ($N=30$ full run at $\lambda=50$: perimeter 174.2 vs 117.8 for seeded- $\lambda=2.1$, +48%). Hence: `init_method`: `seeded` + moderate λ .

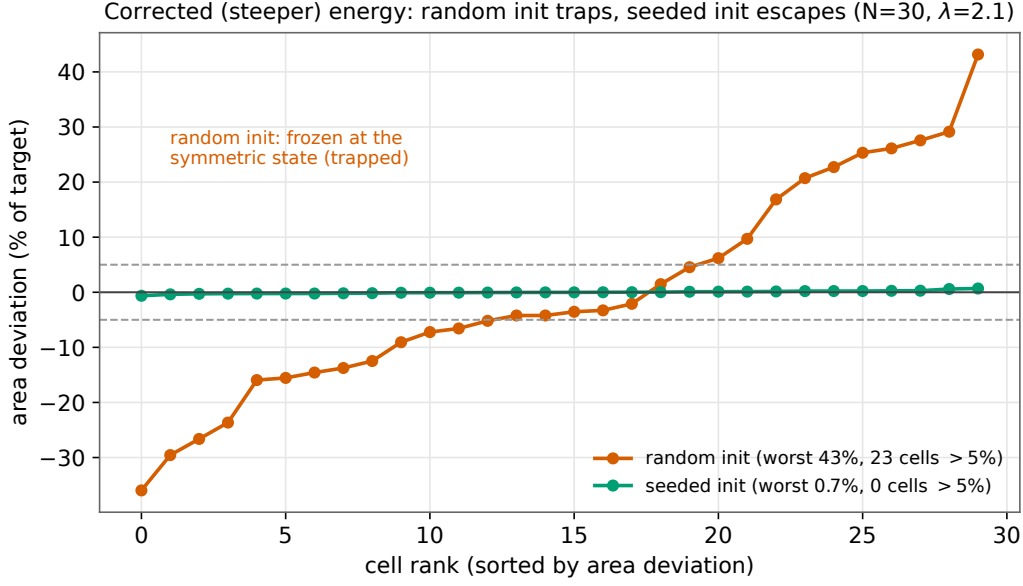


Figure 3: Corrected-energy $N=30$ head-to-head ($\lambda=2.1$, identical but for `init_method`). Random init (vermilion) is frozen at the symmetric state — 23 cells beyond $\pm 5\%$, worst 43%; seeded init (green) is tight within $\pm 1\%$.

4 Conclusion

The high- N runt of report 01 was, in large part, an artifact of a mis-discretized energy rather than a fundamental limit at $N=100$. Two consequences of the $\sim 25\times$ -too-weak well drove it: the diffuse “halo” a cell parks its mass in was priced far too cheaply, and the crispness weight λ was rendered inert. Correcting the well ([docs/reference/phase1.energy_discretization_bug.md](#)) removes both. With the corrected energy, seeded initialization, and a *moderate* $\lambda=5.1$, the $N=100$ torus partition is uniformly equal (worst cell 0.8%) and Phase 2 refines it cleanly (perimeter -13.6% , feasible throughout), yielding the exported production partition. Neither the energy fix alone nor λ alone suffices — they are complementary, and seeded init is now required to avoid the corrected energy’s symmetric trap. No hard discrete-territory or crispness *constraint* (the front-runner in report 01’s ranking) was needed: the soft λ penalty is enough once the well is priced correctly. The standing explanation, with the residual near-convergence migration churn noted, is in [docs/reference/winner_take_all_partition_gap.md](#) §8.